# Web Analytics Final Project

Project Ideas, Feasibility Ranking, and Recommended Direction

This document consolidates the project-planning analysis for the Web Analytics final project. It is designed around the course requirements and the project instructions provided by the professors: define a clear question, collect web data through scraping or APIs, apply data-analysis techniques, present results visually, and review the state of the art.

**Core strategy: choose a project that is technically substantial but still simple enough that every group member can understand the full pipeline and explain the code during the final exam.**

***Recommended shortlist:*** *1) Hacker News success & discussion networks, 2) GitHub project health & contributor networks, 3) Wikipedia attention dynamics.*

# 1. What the Project Needs to Demonstrate

The project instructions require four core elements: web data collection using scraping or APIs; one or more data-analysis techniques studied in the degree; data visualization; and a state-of-the-art review. Graph analysis and an interactive dashboard are desirable but not mandatory.

- **Problem definition:** begin with a specific question or hypothesis.
- **Data source:** use a web source that can be accessed reliably through an API or scraper.
- **Analysis:** apply meaningful statistical, machine-learning, NLP, or graph techniques rather than only descriptive charts.
- **Visualization:** present findings clearly; a Streamlit dashboard is a strong optional deliverable.
- **State of the art:** identify related studies, tools, or previous approaches and explain how the project differs.

The final exam is especially important for project design: the professors can ask about the dataset, analysis techniques, visualization choices, and the exact code used to implement specific steps. Therefore, unnecessary technical complexity can become a liability.

# 2. Overall Ranking

| Rank | Project | Coding | Analytics | Presentation | Risk | Overall |
|---|---|---|---|---|---|---|
| 1 | GitHub Project Health & Contributor Networks | Medium | Excellent | Excellent | Low-Med | 9.4/10 |
| 2 | What Makes a Hacker News Post Successful? | Easy-Med | Excellent | Excellent | Very low | 9.3/10 |
| 3 | Wikipedia Attention & Editing Around Major Events | Easy-Med | Very good | Excellent | Very low | 9.0/10 |
| 4 | Steam: What Drives Positive Game Reviews? | Medium | Very good | Excellent | Low | 8.7/10 |
| 5 | Online News Agenda & Topic Networks | Medium | Excellent | Very good | Medium | 8.4/10 |
| 6 | E-commerce Price & Product Analysis | Med-High | Good | Very good | High | 7.7/10 |

**Best overall fit:** GitHub and Hacker News. Both naturally connect web data acquisition, statistical analysis, graph analytics, and visualization.

# 3. Idea 1 - GitHub Project Health & Contributor Networks

**Core question:**

> *What makes an open-source GitHub project healthy, and how does its contributor network relate to project activity and success?*

## Data

Use the GitHub API to collect data for a controlled set of repositories, ideally within one domain such as AI/Data Science or popular Python open-source libraries.

- Stars and forks
- Issues and pull requests
- Contributors and commits
- Repository age and programming language
- Issue closing times
- Pull-request closing or merging times

- Contributor activity
- Repository topics when useful

### Graph component

Create a bipartite network in which contributors connect to repositories, or construct developer-to-developer networks based on shared contribution activity.

```
Contributor -> Repository or Developer <-> Developer
```

Possible graph metrics:

- Degree centrality
- Betweenness centrality
- PageRank
- Connected components
- Community detection
- k-core
- Network density
- Degree distribution

### Strong analytical hypothesis

Are healthier repositories associated with more decentralized contributor networks?

The analysis could test whether decentralization is related to faster issue resolution, greater commit activity, more stars, or longer project survival.

### Analysis techniques

- **Graph analysis:** centrality, communities, k-core, contribution concentration.
- **Statistics:** correlations, distributions, hypothesis testing.
- **Machine learning:** interpretable regression or clustering to classify repository archetypes.

An example model could estimate project popularity from contributors, network density, issue response time, project age, and other repository characteristics.

### Dashboard concept

- **Overview:** repositories, contributors, stars, issues, PR activity.
- **Network:** selectable repository, interactive contributor network, central users, communities.
- **Project health:** issue resolution, contribution concentration, commit activity.
- **Comparison:** select Project A vs Project B.

### Main risk

GitHub API rate limits and pagination. This is manageable, and it also provides a legitimate challenge to discuss in the final presentation.

**Confidence: 0.95** - Excellent course fit and high analytical value without excessive implementation complexity.

## 4. Idea 2 - What Makes a Hacker News Post Successful?

**Core question:**

> *What factors determine whether a Hacker News submission becomes highly successful?*

This project has one of the best grade-potential-to-workload ratios because the research question is immediately understandable while the analysis can still be sophisticated.

## Data

Use the Hacker News public API and collect a defined period, for example all stories during three months.

- Post ID and title
- URL and linked domain
- Author
- Timestamp
- Score
- Number of comments
- Post type
- Comment tree
- Commenter IDs

## Potential explanatory features

- **Content:** title length, question title, Show HN, Ask HN, keywords, topic, linked domain.
- **Temporal:** hour of day and day of week.
- **Author:** historical activity and prior average score.
- **Network:** commenter count, comment-tree depth, interaction structure.

## Outcome definition

```
Success = 1 if the post is in the top 10% by score; otherwise Success = 0
```

An interpretable classifier such as logistic regression or a random forest can then be used to identify which factors are most strongly associated with high engagement.

## Graph component

Construct a user interaction network where nodes are Hacker News users and edges represent interaction within the same discussion.

- Centrality
- Communities
- Degree distribution
- k-core
- Influential users

A compelling secondary question is: **Do successful discussions have different network structures from ordinary discussions?**

## NLP component

Keep NLP interpretable and manageable rather than using an unnecessarily complex language model.

- TF-IDF
- Sentiment
- Topic clustering
- Keyword analysis

### Dashboard concept

- **Overview:** total posts, comments, users.
- **Attention patterns:** score by posting hour, weekday, domain, topic.
- **Predicting success:** coefficients or feature importance.
- **Network:** user interaction communities and central actors.
- **Post explorer:** select a post and inspect its comment tree.

**Confidence: 0.96** - Probably the safest technically strong project of the shortlist.

## 5. Idea 3 - Wikipedia Attention Around Major Events

**Core question:**

> *How do major real-world events affect Wikipedia attention and editing behaviour?*

### Possible event categories

- Elections
- Product releases
- Sporting events
- Corporate events
- Movie releases
- Scientific events

### Data and analysis

Wikipedia APIs can provide page views, revisions, editors, article links, and metadata. The project could compare page-view and editing dynamics around an event time.

- Peak page-view delay
- Duration of attention
- Editing intensity
- Editor concentration
- Before/after comparisons
- Time-series clustering if useful

### Graph extension

Construct an article-link network around a topic and calculate PageRank, centrality, and communities to study information structure as well as attention.

**Confidence: 0.93** - Reliable APIs, low operational risk, strong time-series visualization, and clear academic framing.

## 6. Idea 4 - Steam: What Drives Positive Game Reviews?

**Core question:**

> *Which characteristics are associated with positive user reception on Steam?*

- Price
- Genres
- Release date

- Developer and publisher
- Review score
- Review text
- Positive/negative label
- Playtime where available
- Tags

### Possible analysis

- **NLP:** sentiment and complaint-topic analysis.
- **Machine learning:** predict positive vs negative reviews.
- **Product analysis:** estimate associations between review score, price, genre, age, and review volume.

The main weakness is that graph analytics are less natural here. There is no need to force a network analysis if it does not improve the research question.

**Confidence: 0.89** - Strong and understandable, particularly if the group is interested in gaming.

## 7. Idea 5 - Online News Agenda & Topic Networks

**Core question:**

> *How does online news coverage differ between publishers in topic selection, timing, and framing?*

Collect RSS feeds and/or article data from several publishers and analyse topics, publication frequency, headline sentiment, named entities, article similarity, and timing.

### Network possibility

Build a publisher-to-topic bipartite graph and use community detection to identify which publishers cover similar subject clusters.

### Risk

Multiple publishers create more scraping and cleaning work, and extraction reliability may vary from site to site.

**Confidence: 0.84** - Academically strong, but more implementation risk than the top three.

## 8. Idea 6 - E-commerce Price Intelligence

**Core question:**

> *How do prices, discounts, inventory, and product characteristics differ across categories and sellers?*

Potential analyses include category pricing, discount behaviour, price distributions, rating-price relationships, brand positioning, and product clustering.

### Main drawback

Commercial websites often use anti-bot protections, dynamic rendering, changing layouts, or restrictive access conditions. The project can become more about defeating access friction than doing analysis.

**Confidence: 0.78** - Good topic, but unnecessarily high operational risk compared with API-first alternatives.

## 9. Projects I Would Avoid

I would avoid building the project primarily around difficult or unstable access to platforms such as Instagram, TikTok, LinkedIn, Google Maps, Booking/TripAdvisor, or X/Twitter.

The issue is not that these topics are uninteresting. The issue is that access restrictions, anti-bot systems, privacy constraints, or expensive APIs can consume time that would be better spent on analysis.

Also avoid the workflow: *scrape something first and then look for insights*. The better sequence is:

```
Question -> Data -> Analysis -> Answer
```

# 10. Recommended Technical Architecture

```
WEB
|
API / Scraper
|
Raw JSON / CSV
|
Cleaning pipeline (Pandas / Python)
|
Statistics + ML + Graph Analysis
|
Results
|
Streamlit Dashboard
|
Conclusions
```

This is enough. Adding Spark, Kubernetes, neural networks, vector databases, or LLM agents would increase complexity without necessarily improving the project.

# 11. Recommended Analysis Structure

## Layer 1 - Descriptive

What does the ecosystem look like? Use distributions, trends, counts, and comparisons.

**Confidence: 0.99**

## Layer 2 - Explanatory

Which variables are associated with the outcome of interest? Use correlation, hypothesis testing, regression, or clustering.

**Confidence: 0.96**

## Layer 3 - Structural

How are participants or entities connected? Use network analysis.

**Confidence: 0.94**

Together, these layers create a natural presentation sequence: **what -> why -> structure -> conclusion**.

# 12. Top-Two Requirement Fit

| Requirement | GitHub | Hacker News |
|---|---|---|
| Web data collection | GitHub API | HN API |
| Significant dataset | Yes | Yes |
| Data analysis | Yes | Yes |
| Graph analytics | Excellent | Excellent |
| ML / statistics | Yes | Yes |
| Visualization | Yes | Yes |

| Requirement | GitHub | Hacker News |
|---|---|---|
| Interactive dashboard | Yes | Yes |
| State of the art | Strong literature | Strong literature |
| Explainable code | Yes | Yes |
| Clear research question | Yes | Yes |
| Interesting demo | Yes | Yes |

# 13. Suggested Pitch Wording

## GitHub

> *Understanding Open-Source Project Health Through GitHub Activity and Contributor Networks*

We aim to investigate whether the structure of developer collaboration networks and repository activity can help explain differences in open-source project health and popularity. We will collect repository, contributor, issue and pull-request data using the GitHub API, analyse repository characteristics and contributor networks using statistical and graph-analysis techniques, and present the results through an interactive dashboard.

## Hacker News

> *What Makes Content Successful on Hacker News? A Web Analytics Study of Content, Timing and User Networks*

We aim to identify the factors associated with highly successful Hacker News submissions. We will collect submission, comment, and user data through the Hacker News API and analyse textual, temporal, and network characteristics of posts and discussions. We will investigate which features are associated with engagement and how discussion-network structure differs between highly successful and ordinary submissions, presenting the results through an interactive dashboard.

The Hacker News formulation is slightly easier to explain immediately because the main research question is obvious to any audience.

# 14. Final Recommendation

The optimal project is not the most technically exotic one. It is the one that lets the group tell a clean analytical story:

```
We asked X.
We obtained Y from the web.
We used techniques A, B, and C.
We discovered Z.
Here is the evidence.
```

**Recommended shortlist:**

1. Hacker News success & discussion networks
2. GitHub project health & contributor networks
3. Wikipedia attention dynamics

The next step should be to design these three finalists at pitch-level detail: exact research questions and hypotheses, API endpoints and fields, expected dataset size, planned analyses, graph construction, dashboard screens, state-of-the-art starting points, workload split, and a 2-3 week implementation plan. Then compare them by grade potential versus workload.

# 15. Confidence Review

| Dimension | Score | Assessment |
|---|---|---|
| Logic | 0.95 | The leading ideas naturally connect data collection, analysis, networks, and visualization. |
| Factual fit | 0.97 | They satisfy the mandatory project requirements and strongly match the course themes. |
| Feasibility | 0.92 | Both can be implemented primarily in Python with structured APIs; access limits are manageable. |
| Presentation quality | 0.96 | Both have one-sentence research questions and strong dashboard/network demos. |
| Weighted confidence | 0.95 | Very strong basis for selecting and pitching a final project. |

**Source basis:** UC3M Web Analytics course description and the uploaded Final Project instructions PDF. The uploaded brief states the mandatory use of scraping or APIs, analysis techniques, visualization, and a state-of-the-art review, while identifying graph analysis and dashboards as desirable additions.